



The Industrial revolution. The birth of true pollution.

that drastic in scale. Supposing someone discovers a battery that is twice as efficient as current designs, we're not going to get devices that last twice as long; we'll get devices that are twice as powerful. Screen sizes will increase; you'll start seeing 8-core or 16-core processors, 4K playback on smartphones and so on and so forth. Very few will actually consider the fact that they may not need more, that the power they have now is actually sufficient to accomplish their tasks and that maybe we should give the world a breather, maybe enjoy 48hr battery life on a four inch screen for once. The only method we are finding to deal with this increased demand is, paradoxically, by trying to find more efficiency and even more fuel somewhere. We want a fuel that is more efficient to store and transport. One that stores the largest quantity of power in the minimum amount of space and one that is also available in abundance. The only fuel that satisfies this criteria is oil. It's available in massive quantities, it's a liquid and is hence, easy to store, transport and distribute. The power output is many times greater in magnitude than any other form of energy currently at our disposal and engines that harness this power can vary in scale from the miniature ones that power model aircraft to the humungous engineering marvels that are used in the generators of power plants.

How big is oil?

Well, this is a tough one. Oil is huge, so huge, that it's actually difficult for us to comprehend without some sort of analogy.

So here are some facts (and analogies):

- Oil, by itself, is responsible for 1/3rd of the entire world's energy supply.
- Oil effectively powers 100% of our transportation. Without oil, the world will literally grind to a halt. This includes electric cars, because a lot of power stations still use oil and fossil fuels.
- The USA spent over \$100 billion on the Iraq war, the bulk of which was the sheer cost of the oil consumed during the conflict.
- About 40% of all seaborne cargo is oil. Also, the quantity of oil passing through the world's oceans every year is greater than the entire known marine biomass. There is literally more oil than fish in the sea.
- If you take the by-products of oil drilling and purification into account as well, including natural gas, then more than half the world's energy is produced by oil.
- Oil and natural gas pipelines in the US alone can make the trip to the moon and back seven times over.
- Oil is everywhere, 2010 figures peg the daily oil production at over 80 million barrels. We are literally using it up as fast as we're getting it out of the ground and even at this alarming rate of utilization, we're still expected to have enough oil to last us decades.
- It will take the world almost a hundred years to retool itself to make use of an energy source that is not, or very dissimilar to, oil.

A world without oil

If we were to imagine a world without oil, the first major impact will be to transportation. There is absolutely no way that we would have anything like supersonic aircraft, in fact, air travel except in something like hot air balloons, would be impossible.

The entire race to get to the skies was defined by the need for power. The Wright brothers managed to get to the air with a spindly little aircraft, but only when the power was enough to keep that aircraft in the air. NASA is experimenting with solar powered aircraft, but even now that speed is nothing, relative to what we can get from burning oil and the only possible use that can be seen for it is in terms of surveillance and maybe as a flying signal tower.

All your Ferraris and Lamborghinis would simply not exist. Sure, you've got electric cars that go at 150mph now, but where do you think that electricity is coming from? Sparing that much power for a 2hr drive would be something that billionaires would consider; it would not be an everyday occurrence.

If these above facts are not staggering enough, consider this: Given the scale at which oil has seeped into our everyday lives, any so-called "green" energy that we are working on right now, will not even make a minor dent in oil utilization and extraction. Sure, a massive solar plant will give us more power, but we'll also soak up that power as fast as it's produced. While larger countries can afford to spend some money on experimental power sources, developing nations like ours can't even imagine doing something like that. Despite the high price of oil, it's still more efficient to buy it by the millions of barrels than invest in large-scale "alternate energy" plants.

The blame game

Are fossil fuels really to blame for this "global warming" and "environmental pollution" that we see all around us? Yes, but only indirectly. It's we who are responsible for using that much oil, that much power and as a consequence of that, generating that much CO₂ and pollution. We're asking for that much power and we're taking it, but without shouldering the responsibility that that power entails.